

Proof H is Sequentially Consistent: same as Proof H is Linearizable expect no need to proof (3).

Proof H is NOT Linearizable: Proof by contradiction. Assume H is linearizable. As H is a complete history (without pending invocations), this means there exists a history S such that:

- (a) S is equivalent to H ;
- (b) S is a legal sequential history; and
- (c) $\rightarrow_H \subseteq \rightarrow_S$

As H is a sequential history and thus $\rightarrow_H$ is total, from (c) we know that $S = H$. On the other hand, we know H is not a legal history (because ...), and thus (since $S = H$) we know S is not a legal history. This, however, leads to a contradiction as in (b) we assumed that S is a legal history.